

What Explains a Pitcher's Value?

Active spin, velocity, and movement as predictors of WAR in Major League Baseball

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Research Question and Data

This report examined how four-seam fastball active spin percentage, velocity, and pitch movement related to pitcher value, measured by Wins Above Replacement (WAR), across the 2021-2025 MLB seasons. Rather than focusing on one player or one season, the analysis used a five-year sample to compare pitchers with consistent Statcast records.

Pitching data came from Statcast and included active spin percentage, average four-seam fastball velocity, horizontal movement, and vertical movement. WAR data came from Baseball Reference and was averaged across the same five seasons. The Statcast data included 93 pitchers with complete records across all five seasons; 92 pitchers matched successfully to the WAR data used in the regression model.

Method

Each pitcher's four-seam fastball metrics were averaged across the five seasons. Horizontal and vertical movement were analyzed separately, and their combined movement was used to classify pitchers into profile groups. Pitchers were labeled high or low for active spin, velocity, and movement based on the median value for each metric. These labels produced eight profiles, such as HHH, HHL, and LLL, that allowed the analysis to compare combinations of traits rather than isolated variables.

The analysis used descriptive statistics, profile comparisons, correlations, visualizations, and a multiple linear regression model. The regression tested whether active spin percentage, velocity, horizontal movement, and vertical movement predicted average WAR when all four variables were included together.

Results

The descriptive statistics showed that the sample consisted mostly of established MLB pitchers with similar fastball characteristics. Average active spin percentage was 88.7%, average velocity was 93.9 mph, average horizontal movement was 7.6 inches, and average vertical movement was 15.6 inches. Average WAR was 1.13 across the matched sample.

The correlations between individual pitch traits and WAR were weak. Active spin percentage had a small negative relationship with WAR ($r = -0.08$, $p = 0.452$). Velocity had a small positive relationship ($r = 0.09$, $p = 0.409$). Horizontal movement ($r = -0.17$, $p = 0.100$) and vertical movement ($r = -0.14$, $p = 0.196$) were negative but not statistically significant.

The profile comparison produced a similar pattern. The highest average WAR appeared in the LHL profile, followed by LLL. Several pitchers with high values across the measured traits did not produce the highest WAR totals. This pattern suggests that combinations of fastball traits did not explain pitcher value by themselves in this sample.

Regression Findings

The multiple regression model was not statistically significant overall, $F(4, 87) = 0.95$, $p = 0.437$, and explained about 4.2% of the variation in average WAR. None of the four predictors had a statistically significant coefficient. The adjusted R-squared value was slightly below zero, which means the model did not improve prediction after accounting for the number of predictors.

Predictor	Coefficient	p-value	Interpretation
Active spin %	-0.009	0.716	Not significant
Velocity	0.010	0.905	Not significant
Horizontal movement	-0.032	0.613	Not significant
Vertical movement	-0.038	0.628	Not significant

Conclusion

The results indicate that four-seam fastball active spin percentage, velocity, and movement did not meaningfully predict pitcher WAR in this five-year MLB sample. Visual patterns suggested that some pitchers with stronger fastball traits produced higher WAR, but the correlations and regression model did not support a reliable statistical relationship.

These findings do not mean that velocity, spin, and movement lack value. They show that WAR depends on more than fastball traits alone. Command, pitch mix, durability, role, sequencing, defense, and run prevention context likely affect pitcher value in ways this model did not capture. At the MLB level, where most pitchers already have elite physical tools, those broader performance factors may separate higher-WAR pitchers from lower-WAR pitchers.

Future research could extend this framework by separating starters and relievers, adding innings pitched or strikeout and walk rates, and comparing MLB results with college or high school pitchers. A lower-level sample may show stronger relationships because pitch traits vary more widely before players reach the major leagues.